25901151 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Discipline Specific Core Course (DSC) - MG4DSCCHE202 - FUNDAMENTALS OF PHYSICAL CHEMISTRY

(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple choice questions

Answer all questions

Each question carries **1** marks

1. In a BCC unit cell, the atom at the body center contributes [U] / [CO1]

   a). 1/8 b). 1

   c). 1/2 d). 1/4

2. Which among the following is a good adsorbent? [An] / [CO2]

   a). Glass b). Charcoal

   c). Water d). Rubber

3. Who invented galvanic cell? [K] / [CO4]

   a).Michael Faraday b). Luigi Galvani and Alessandro Volta

   c). Humphry Davy d). Walther Nernst

4. The forbidden energy gap is maximum in [U] / [CO1]

   a). conductors b). semiconductors

   c). Insulators d). Superconductors

5. Name the process in which the sol particles migrate toward the positive electrode during electrophoresis is called [K] / [CO3]

a).cataphoresis b).anaphoresis

c).electro-osmosis d).sedimentation potential

6. Which statement is correct regarding the Kohlrausch's law? [U] / [CO4]

a). Conductance depends on concentration	b). Molar conductance at infinite dilution is the sum of individual ionic conductances	c). Conductance is independent of ions	d). Conductance decreases exponentially with dilution

7. Particles of suspension can be separated by [U] / [CO3]

a). Dialysis b). Ultrafiltration

c). Ordinary filtration d). Diffusion

8. If the same current flows through $CuSO_4$ and $AgNO_3$ solutions, which will deposit more mass (Equivalent weight of Silver = 107.87 g; Copper = 31.77 g). [A] / [CO4]

a). Copper b). Silver

c). Both equal d). Cannot be predicted

9. Molar conductance (ΛM) is defined as the conductance of [U] / [CO4]

a). 1 liter of electrolyte

b).solution between electrodes 1 cm apart

c).solution containing 1 gram equivalent of electrolyte

d).solution containing 1 mole of electrolyte

10. The electrochemical series is arranged according to: [K] / [CO4]

a).Atomic radius b). Standard electrode potentials

c). Atomic mass d). Electrical conductivity

[1x10 = 10]

Part B

Short Answer Questions

Answer any **4** questions

Each question carries **5** marks

11. Explain how crystal systems are classified based on axial lengths and angles. [U] / [CO1]

12. Discuss briefly on a) electrodialysis b) ultrafiltration. [U] / [CO3]

13. Mention any four applications of conductance measurements. [An] / [CO4]

14. What are temporary magnets? Why is soft iron preferred for making temporary magnets? [U] / [CO1]

15. Discuss the nature of graph obtained in the conductometric titration of strong acid versus weak base by taking suitable examples. [U] / [CO4]

16. Define conductors. Explain the effect of temperature on the conductivity of conductors. [U] / [CO1]

[5x4 = 20]

Part C

Short Essay Questions

Answer any **2** questions

Each question carries **10** marks

17. What are emulsions? Discuss the applications of emulsions in medicine and cleansing action of soap. [A] / [CO2]

18. Write a short note of the following a) Hardy-Schulz rule b) electrical double layer [U] / [CO2]

19. Explain the principle and working of redox electrodes using a platinum electrode. [A] / [CO4]

20. Explain the principle, instrumentation, and working of potentiometric titration. Discuss the role of indicator electrode and reference electrode with suitable examples. [K] / [CO4]

[10x2 = 20]